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ORIGINAL ARTICLE

Inference of Functional Divergence Among Proteins When the Evolutionary Process is Non-stationary

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Abstract Functional shifts during protein evolution are expected to yield shifts in substitution rate, and statistical methods can test for this at both codon and amino acid levels. Although methods based on models of sequence evolution serve as powerful tools for studying evolutionary processes, violating underlying assumptions can lead to false biological conclusions. It is not unusual for functional shifts to be accompanied by changes in other aspects of the evolutionary process, such as codon or amino acid frequencies. However, models used to test for functional divergence assume these frequencies remain constant over time. We employed simulation to investigate the impact of non-stationary evolution on functional divergence inference. We investigated three likelihood ratio tests based on codon models and found varying degrees of sensitivity. Joint effects of shifts in frequencies and selection pressures can be large, leading to false signals for positive selection. Amino acid-based tests (FunDi and Bivar) were also compromised when several aspects of the substitution process were not adequately modeled. We applied the same tests to a core genome "scan" for functional divergence between light-adapted ecotypes of the cyanobacteria *Prochlorococcus*, and carried out gene-specific simulations for ten genes. Results of those simulations illustrated how the inference of functional divergence at the genomic level can be seriously impacted by model misspecification. Although computationally costly, simulations motivated by data in hand are warranted when several aspects of the substitution process are either misspecified or not included in the models upon which the statistical tests were built.

Keywords Power · Accuracy · Functional divergence · Positive selection · Non-stationary evolution

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Introduction

Statistical methods can provide valuable information about adaptive genetic variation and the process of functional divergence (Bielawski and Yang 2004; Gaucher et al. 2002). Many tests are derived from the concept that the substitution rate at a site is inversely related to its functional constraint (Kimura 1983). Therefore, a measurable shift in the nonsynonymous substitution rate over some period of time is commonly interpreted as evidence of functional divergence (e.g., Gu 1999; Knudsen and Miyamoto 2001; Susko et al. 2002). A variety of strategies have been developed for detecting shifts in the nonsynonymous rate. These can be grouped into two broad categories: (1) methods based on a nonsynonymous rate scaled by the inverse of the synonymous rate, and (2) methods based on the absolute rate of nonsynonymous substitution.

Scaling the nonsynonymous rate (d_N) by the synonymous rate (d_S) yields a useful index, $\omega = d_N/d_S$, of the intensity of natural selection. Here, the rate of synonymous

change represents the rate of gene evolution prior to the impact of selection on the protein. When a protein is subject to positive Darwinian selection, the nonsynonymous rate can exceed the synonymous rate yielding $\omega > 1$. To detect such cases, the ω ratio was formulated as an explicit parameter of a Markov model of codon evolution (Goldman and Yang 1994; Muse and Gaut 1994). Among the more recent developments are the branch-site models, which permit sites to have a unique ω along a predetermined lineage within a phylogeny (Yang and Nielsen 2002; Bielawski and Yang 2004; Zhang et al. 2005; Murrell et al. 2012). These models aid detection of functional shifts by enabling explicit tests for sites having changes in selection pressure among branches or subtrees.

Alternatively, functional divergence can be detected through differences in the absolute rate of nonsynonymous substitution. In this approach the rate is an explicit parameter in a model of amino acid evolution, where it is referred to as the "evolutionary rate." A shift in rate between subtrees is commonly referred to as "Type-I divergence." There are several methods for detecting Type-I divergence (e.g., Gu 1999; Knudsen and Miyamoto 2001; Susko et al. 2002) and they share the same overall approach. First, two subtrees are specified a priori and the rate at each site is estimated separately for each subtree. These rates are then used to measure the magnitude of divergence between the subtrees. Methods differ in how rate shifts are quantified, how they are statistically tested, and how other model parameters are treated. Regardless of those differences, sites inferred to have a rate-shift among subtrees are interpreted as having undergone a shift in functional constraint.

In addition to Type-I divergence, a difference in amino acid frequencies between subtrees has been interpreted as signal for functional divergence (Type-II divergence). Indeed, it is not unusual for both amino acid composition (or codon usage) and evolutionary rates to change at the same point in time as protein function (e.g., Kettler et al. 2007; Zhang et al. 2005). However, current methods for detecting Type-I divergence assume a single Markovian process accurately represents all aspects of evolutionary change and that this process is at equilibrium (i.e., stationarity). Indeed, when the task is phylogenetic inference, inadequate modeling of shifts in codon bias or amino acid composition is a known cause of analytical artifacts (Lake 1994; Mooers and Holmes 2000; Chang and Campbell 2000; Inagaki and Roger 2006). Although nucleotide models that permit non-stationary composition provide a much better fit to some datasets (Yang and Roberts 1995; Galtier and Gouy 1998), no parallel models exist at the codon level. Model-based methods for detecting functional divergence have proved valuable, and they are now used in a wide variety of settings. However, we do not understand the extent to which this level of model misspecification could lead to analytical artifacts and, ultimately, false biological conclusions.

We used a series of simulations to explore the effects of non-stationary frequencies on the inference of functional divergence. Specifically, we addressed: (1) the impact of non-stationary codon frequencies on tests derived from codon models, (2) the combined impact of non-stationary frequencies and changing selection pressure on those same tests, and (3) the power and reliability of tests derived from amino acid models when frequencies are non-stationary. We found that tests formulated at both the codon and amino acid level can yield false signal for altered selection intensity when several aspects of the model are inadequate. To investigate the impacts of these limitations on the analysis of real data, we also analyze the core genome of the cyanobacteria *Prochlorococcus*, which is known to have non-stationary sequence evolution.

Methods

Codon Level Tests

We employed four codon models that permit three different likelihood ratio tests (LRTs) for a shift in selection pressure (Yang and Nielsen 2002; Zhang et al. 2005). The LRTs were based on *site models* named M1a and M3, and *branch-site models* named Model A and Model B. The LRTs were:

Test 1: M1a versus Model A (Zhang et al. 2005).

Test 2: Model A ($\omega_2 = 1$) versus Model A ($\omega_2 > 1$) (Zhang et al. 2005).

Test 3: M3 versus Model B (Yang and Nielsen 2002).

For each LRT, the test statistic was two times the difference in log-likelihood between the involved pair of models ($2\Delta\ln L$). Tests 1 and 3 have an asymptotic χ^2_2 distribution. Although the asymptotic distribution for Test 2 is $1/2\chi^2_1 + 1/2\chi^2_0$, Zhang et al. (2005) recommend the more conservative χ^2_1 distribution to increase robustness to model violations. We employed both qa and the asymptotic mixture distribution.

For branch-site models, the so-called foreground branches are specified a priori and represent periods of evolutionary history where a fraction of sites have experienced a change in selection pressure as compared to the other (background) branches. Model A has four categories of sites. Two categories of sites are comprised entirely of background branches, with one category having only purifying selection ($\omega_0 < 1$) and the other having only neutral evolution ($\omega_1 = 1$). The remaining two categories are comprised of sites having both background branches

(with $\omega_0 < 1$ or $\omega_1 = 1$) and foreground branches constrained to have positive selection ($\omega_2 > 1$). The null model for Test 2 is a modified version of Model A. It does not allow for positive selection in the foreground branch ($\omega_2 = 1$). Model M1a permits no changes between the foreground and background branches and no positive selection. M1a is comprised of two site categories with one category having only $\omega < 1$ and the other having only $\omega = 1$. Tests 1 and 2 use these models to test for sites subject to *positive selection* in foreground branches. M3 consists of two categories of sites that have constant ω over branches. No constraints are placed on the ω's of M3. Model B is similar to Model A in the way that it allows selection pressure to change between foreground and background branches, but its ω's are unconstrained (as in M3). Test 3 uses this pair of models to detect *any shift in selection pressure* in the foreground branches.

Amino Acid Level Tests

We employed two methods to test for amino acid rate shifts: Bivar (Susko et al. 2002) and FunDi (Gaston et al. 2011). Bivar separately measures evolutionary rates for subtrees, which are defined a priori. For a given site, i, the evolutionary rate (r_i) is measured using a conditional mode estimate; i.e., the rate with the greatest probability given the sequence data and phylogeny. We compared rate differences between subtrees (denoted $r_{i\mathrm{A}}$ and $r_{i\mathrm{B}}$) by using three summary statistics described by Susko et al. (2002): (1) *arsum*—the sum of the absolute values of $r_{i\mathrm{A}}$–$r_{i\mathrm{B}}$, (2) *alrsum*—the sum of the absolute values of the log of ratio $r_{i\mathrm{A}}/r_{i\mathrm{B}}$, and (3) *abrsum*—the values of *arsum* scaled by the sum of the conditional mode estimates.

$$arsum = \sum_i |r_{i\mathrm{A}} - r_{i\mathrm{B}}|$$

$$alrsum = \sum_i |\log(r_{i\mathrm{A}}/r_{i\mathrm{B}})|$$

$$abrsum = \sum_i |r_{i\mathrm{A}} - r_{i\mathrm{B}}|/(r_{i\mathrm{A}} + r_{i\mathrm{B}})$$

Significance was evaluated via a parametric bootstrap (Susko et al. 2002).

FunDi uses a likelihood-based mixture model to detect site-specific rate-shifts. The model is a two-component mixture of (1) a standard evolutionary model, where substitution rates are constant throughout the phylogeny, and (2) a model that permits different rates in the two subtrees. We developed an LRT based on this model to formally test the hypothesis that a fraction of sites have different rates in the subtrees. The test-statistic was two times the difference in log-likelihoods ($2\Delta\ln L$) between a null model that has equal rates in subtrees and the FunDi mixture model. This LRT will not be χ^2 distributed due to the multiple-parameter boundary problem (e.g., case 8 in Self and Liang 1987). Hence, we used a parametric bootstrap for the FunDi LRT.

Parametric Bootstrapping

Data generated for parametric bootstrapping was based on a maximum likelihood phylogeny obtained by applying RAxML (Stamatakis 2006) to an amino acid alignment. The shape parameters of a gamma model for rate across sites, the proportion of invariable sites, and the branch lengths were estimated via ML from the same alignment by using the codeml program in the PAML package (Yang 2007). The parameter estimates were obtained under the WAG model (Whelan and Goldman 2001) combined with empirical amino acid frequencies. Those values were then used to simulate 100 parametric bootstrap samples with stationary frequencies and no shifts in evolutionary rate (the null model). Each bootstrap sample was then split into subtrees, based on the original alignment, and analyzed with Bivar or FunDi. A null distribution was derived for each test statistic (*arsum*, *alrsum*, and *abrsum* for Bivar, and $2\Delta\ln L$ for FunDi).

Simulation Studies

Using the INDELible simulation software (Fletcher and Yang 2009), we simulated data under 18 different evolutionary schemes. Simulated datasets, each 200 codons in length, were based on the ten-taxon, asymmetric phylogeny shown in Fig. 1a. Zhang (2004) suggested that tree topology could be a factor in false positive rates, but Zhang et al. (2005) could find no difference in those rates between symmetric and asymmetric topologies. We used the 10-taxon asymmetric topology because (1) it was previously used by Zhang et al. (2005) in their initial evaluation of Test 1 and 2, (2) Bay and Bielawski (2011) also used it to investigate recombination-detection under scenarios relevant to functional divergence, and (3) the core-genome tree for our empirical data set is asymmetric. We simulated shifts in codon bias, selective constraints, or both, at a single point in evolutionary history (Fig. 1a). This yielded subtrees with different evolutionary models, called "Type A" and "Type B" (Fig. 1).

Codon bias was modeled according to Aris-Brosou and Bielawski (2006) (See supplementary materials). The parameter "η" provides a simple model for specifying codon usage bias as a function of %G+C at the third codon position (GC3). η ranges from 0 to 1, with the value of $\eta = 0.5$ indicating a GC3 content of $\sim$50 %; i.e., all non-stop codons have equal frequencies. GC3 content, and codon bias, increases or decreases as η departs from 0.5.

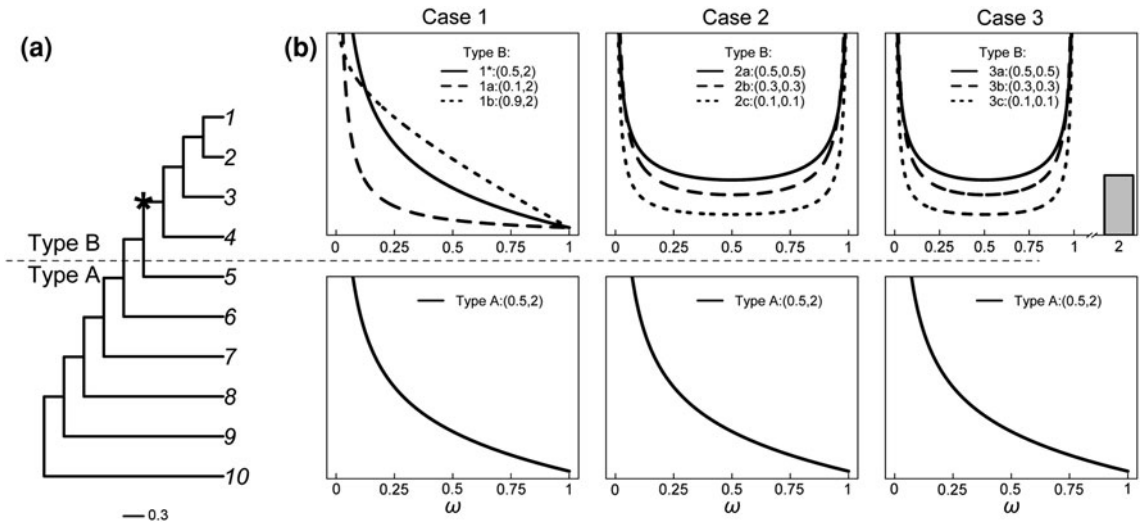

Fig. 1 **a** Phylogenetic tree used to simulate codon sequences. Internal branch lengths in the phylogeny are set to 0.3 nucleotide substitutions per codon, and all other branch lengths are adjusted to achieve rate constancy. When a shift in either selection pressure or codon bias is simulated, it occurs at the node marked by an *asterisk* (*), which defines branches having Types A and B evolution. **b** Distribution of ω for Types A and B evolution. A beta distribution is used for variability in ω among sites, and the shape parameters (p, q) for sub-cases *a–c* are given in legends of the beta distribution plot. A shift in the distribution of ω within the tree is modeled by specifying different shape parameters for Types A and B evolution

We specified phylogenetic shifts in codon biases by setting different η values for Type A and B models.

Shifts in functional constraints were simulated at the codon level by specifying different ω distributions for Type A and B models. These distributions were modeled using the β distribution, which permits a variety of shapes by using its two shape parameters (p, q). For the purpose of generating data sets without positive selection, we split a β distribution into ten discrete categories of equal probability. When positive selection was included in a model, the β distribution was divided into nine categories and an extra site category having $\omega = 2$ was specified for the remaining 10 % of sites.

For all simulation scenarios, the six most deeply branching taxa in the phylogeny were simulated under Type A evolution. For this model, most codon sites are under strong purifying selection (ω <<1) and only few sites are evolving close to neutrality. Specifically, we set $p_A = 0.5$ and $q_A = 2$, which results in a "L-shaped" β distribution (Fig. 1b). The parameter η_A was set to 0.5 (GC3 $\approx$ 50 %).

Type B evolution varied among simulation scenarios. We simulated three basic cases for Type B evolution (Fig. 1b):

Case 1: Most sites are under strong purifying selection. This represents an "easy null" for tests formulated to detect positive selection because there are few sites evolving on the boundary ($\omega = 1$) between the null hypothesis ($\omega < 1$) and the alternative hypothesis ($\omega > 1$).

Case 2: A large proportion of sites evolve close to neutrality. This is considered a "hard null" for tests formulated to detect positive selection because the increased proportion of sites near neutrality ($\omega \approx 1$) are more likely to give a false signal for positive selection.

Case 3 (positive selection): This extends Case 2 by adding a category of sites having evolved under positive selection ($\omega = 2$).

Within each case of Type B evolution, we simulated different "sub-cases", designated via lowercase letters (Fig. 1b). One scenario within Case 1 represents homogenous selection pressure, and is denoted as Case 1*. The design of Cases 1 and 2, and their subcases, matches those used by Bay and Bielawski (2011). Case 3 extends that design by adding positive selection. Taking Types A and B evolution together yields the following scenarios: (1) homogenous selection pressure [Case1*]; (2) small shifts in selection [Cases 1a and 1b]; (3) large shifts in selection [Cases 2a–c]; and (4) large shifts with positive selection [Cases 3a–c]. Each case was simulated both with a phylogenetic shift in codon bias ($\eta_B = 0.1$) and without a shift ($\eta_B = 0.5$). In total, this design yielded 3 cases $\times$ 3 subcases $\times$ 2 models of codon bias = 18 different evolutionary scenarios.

Each of the 18 scenarios was analyzed at both codon and amino acid level. Codon models were fit to data by using PAML (Yang 2007). For branch-sites models, branches under Type B evolution were specified as foreground branches and branches simulated under Type A evolution were background branches. Bivar and FunDi were applied to translated sequences, with the Type A and B branches specified as different subtrees.

Analysis of *Prochlorococcus* Genomic Data

Prochlorococcus marinus (Chisholm et al. 1988) is a marine cyanobacterium that contains both high-light-adapted (HL) and low-light-adapted (LL) ecotypes (Moore et al. 1995; Partensky et al. 1997; West and Scanlan 1999; Hess et al. 2001; Rocap et al. 2002). We analyzed protein-coding genes from a subset of the HL + LL core genome from *Prochlorococcus marinus*. The subset included 585 genes sampled from 12 complete genomes. These genes were selected because their gene-trees had the same basic structure as the organismal tree (see supplementary materials for details about the data processing methodology). All genes in the subset were subjected to analysis of functional divergence with codon Tests 1 and 2, Bivar and FunDi. For the branch-site models, HL and LL branches were specified as foreground and background branches, and for Bivar and FunDi as separate subtrees. We also conducted an analysis in which only the single branch between ecotypes was specified as the foreground branch in the branch-site models.

Results and Discussion

Codon Tests 1 and 3 are More Sensitive to Non-stationary Codon Bias than Codon Test 2

Because Tests 1–3 are based on models assuming that codon frequencies are at equilibrium, we explored how a phylogenetic shift in codon usage impacts their use for inference of selection pressure. The simplest scenarios in our simulation, stationary and non-stationary Case 1*, serve to isolate the impact of codon frequencies. Because selection intensity is homogeneous across the tree in Case 1* the null models should not be rejected at a rate higher than the chosen significance level (5 % in this study). Note that the non-stationary Case 1* does represent Type-II divergence at the amino acid level (see supplementary Fig. S2), but these tests are not intended to detect such divergence.

Results for Case 1* are presented within Table 1. Test 2 performed well in Case 1*, with 0 % false positives when codon usage was both stationary and non-stationary. Tests 1 and 3 yielded a false positive rate greater than the level of the test when codon usage was stationary (9 and 11 %, respectively) and this increased substantially under non-stationary codon usage (27 % for both tests). These results suggest that Test 2 might be preferable for datasets where a shift in codon usage is associated with the foreground branches of the model.

The robustness of Test 2 is not due to the naive use of $df = 1$ as recommended by Zhang et al. (2005), because the false positive rate was 0 % when the appropriate mixture distribution was used. The robustness of this test is likely a consequence of the constraints placed on the parameters of Model A, which serves as the basis for both the null and alternative models of this LRT. In Test 2 the null version of Model A has the ω parameter of the foreground branches fixed at 1 (neutrality) whereas the alternative version is constrained to positive selection ($\omega > 1$). Test 1 and Test 3 differ in that their null models completely exclude a class of sites with a different ω in the foreground. Since the foreground ω is the only model parameter that is adjustable to the foreground branches, it can "soak up" un-modeled variation in frequencies. In sum, Tests 1 and 3 are mistaking codon usage variability among lineages as a signal for a shift in the distribution of selective constraints.

Table 1 Performance of likelihood ratio tests (LRTs) for shifts in selection pressure (ω) along pre-determined branches of a phylogeny: percent replicates with significant LRTs ($p \leq 0.05$)

	Test 1		Test 2		Test 3	
Simulation	S	NS	S	NS	S	NS
Null						
Case 1*	9.0	27	0 (0)	0 (0.5)	10.5	26.5
Small shifts in selective constraints						
Case 1a	0.5	3.5	0 (0)	2 (2)	75	72.5
Case 1b	59.5	92	0 (0)	0 (0)	94	95
Large shifts in selective constraints						
Case 2a	97.5	99	0 (0)	0 (0.5)	94	99
Case 2b	91.5	97	0 (0)	2.5 (5.5)	76.5	96
Case 2c	83.5	91	3.5 (6)	32 (42.5)	72	93.5
Large shifts with positive selection						
Case 3a	97	99	0 (0)	3.5 (5)	97.5	98.5
Case 3b	96.5	99	0 (0)	9 (15.5)	93	99.5
Case 3c	94	94.5	3.5 (6)	48.5 (58)	89	98

For Test 2 results are presented under χ^2_1, and (in parentheses) a mixture $1/2\chi^2_1 + 1/2\chi^2_0$. For Tests 1 and 3, results are based on χ^2_2

Note The LRTs used are: Test 1, M1a versus Model A; Test 2, modified Model A ($\omega_2 = 1$) versus Model A; and Test 3, M3 versus Model B. Case 1* indicates homogenous distribution of selection pressure for all branches in the tree. Case 1* is a null scenario with respect to all three statistical tests. All other cases represent scenarios where the distribution has changed over the course of evolution. Case 3 tests for power for all LRTs and Cases 1(a–b) and Case 2 test for power for Tests 1 and 3. Sub-cases are given in the order of increasing fraction of sites on the upper boundary of the null hypothesis (i.e., $\omega = 1$). In each case, codon usage bias was both stationary (S) and non-stationary (NS)

The Impact of Joint Changes in Codon Usage and Selection Pressure

An unresolved question is to what extent Tests 1–3 are impacted by a joint shift in functional constraints and codon frequencies. Five of our simulation scenarios were characterized by either a small shift (Cases 1a and 1b) or a

large shift (Cases 2a, 2b, and 2c) in selection pressure between foreground and background branches. In none of those cases did the shift involve positive selection; hence they correspond to different null scenarios for the explicit tests for positive selection (Tests 1 and 2). These scenarios are also relevant to the question of codon bias because they were generated with both stationary (S) and non-stationary (N) codon frequencies.

For Test 1, just a small shift in the distribution of selection pressure (Case 1b) caused an increase in the false positive rate (Table 1). Even when codon frequencies are stationary and there was only a slight shift in selection pressure, the false positive rate was as high as 59.5 % (Table 1: Case1b). The addition of non-stationary codon bias to this case caused false positive rates to increase substantially (92 %). It is interesting to note that Case 1a yielded fewer false positives than the null (Case 1*), likely because Case 1a involved a shift to fewer sites evolving close to neutrality (see Fig. 1b), which is the boundary between the null and alternative models of Test 1. The problem of false positives was magnified in Case 2, where there was a larger shift in selective pressures. For Case 2 the overall effect of non-stationary codon bias on Test 1 was small simply because the false positive rate was already unacceptably high (84–98 %) when codon frequencies were stationary. Although our primary concern is with the false positives in the LRTs, and not parameter estimates, it is interesting to note that, the distribution of ω_2 estimates also can be impacted by non-stationary codon frequencies (Table 2). This supports the idea that the unmodeled shift in codon bias is being absorbed by the model through phylogenetic variation in ω_2. Our findings are consistent with a previous study, simulated under very different conditions, which found high false positive rates for Test 1 (40–70 %) depending on topology, branch lengths, and selection pattern (Zhang 2004). A subsequent study suggested that Test 1 is better suited as an LRT for relaxed constraint than positive selection (Zhang et al. 2005). Our results indicate that it is too liberal for this purpose when codon bias is not stationary (Table 1: Case1*).

Test 2 is known to be a more conservative test for positive selection (Zhang et al. 2005). We found that when a small shift in selection pressure was present (Case 1), the false positive rate was very low regardless of whether codon bias was stationary (Table 1). For a large shift in selection pressure (Case 2) combined with a large shift in codon frequencies (via η), the false positive rate was as high as 32 %. However, this increase was due to one subcase (Case 2c), which had a high proportion of sites evolving at, or close to, $\omega = 1$. Indeed, the shape of the beta distribution in Case 2c yields a distribution for ω that is close to a boundary scenario that is well known for generating high false positives in some LRTs (Bao et al. 2008, 50 % $\omega = 1$ and 50 % $\omega = 0$). Thus, Test 2 becomes unreliable only in the very extreme case where a large shift on codon bias is accompanied by a fraction of sites that change from strong purifying selection pressure ($\omega \approx 0$) to nearly neutral evolution ($\omega \approx 1$). Taken together, our results indicate that Test 2 is generally robust as a test for positive selection, but it can suffer increased false positives when there is an extraordinary shift in several aspects of the evolutionary process over the tree.

Table 2 Differences in foreground ω_2 estimates for simulations under stationary (S) and non-stationary (NS) codon bias under codon-level Model A

	Case 1		Case 2		Case 3	
	S	NS	S	NS	S	NS
Median ω	1.0	1.0	1.0	1.0	1.0	1.0
I.Q.R of ω	1.0–1.0	1.0–1.0	1.0–1.0	1.0–1.4	1.0–1.0	1.0–1.6
K.S. test	$P = 0.899$		$P < 0.0001$		$P < 0.0001$	

Note I.Q.R. represents the inter-quartile range (1st–3rd quartiles) of the data. K.S. indicates Kolmogorov–Smirnov test. The K.S. test was used to test the equality of the ω_2 distributions obtained from the stationary and non-stationary cases. Sub-cases were combined for the purpose of testing the stationary versus non-stationary cases

Test 3 is based on M3 and Model B. This test is not an explicit test for positive selection in foreground branches, but rather a test for any change in the intensity of selection pressure (Yang and Nielsen 2002). Because a significant result for Test 3 in Case 1a, Case 1b, or Case 2 should not be interpreted as evidence of positive selection, it is not a false positive. In those cases Test 3 had high power to detect a shift in selection pressure (>72 %). Recall that Zhang et al. (2005) suggested that Test 1 be better employed as a test for a shift in selective constraints. As such, Test 3 is preferable to Test 1 when frequencies are stationary because it has more power to detect small shifts in selective constraints (Cases 1a and 1 b). Test 3 has even greater power under non-stationary codon bias. Because this results from a model violation, it can be considered a systematic error in the direction of the alternative model. Given that the relationship between the assumptions of the model and the true generating process will be unknown for real data, it seems imprudent to rely on systematic errors in the direction of the alternative hypothesis as means of achieving good power.

Case 3 (a–c) is relevant to assessing power to correctly reject the null model when positive selection is present (Table 1). Tests 1 and 3 had high power in Case 3. Unfortunately, the only test for selection that was robust to non-stationary frequencies under the null, Test 2, had low power. The best performance of Test 2 is under

non-stationary conditions in Case 3, but the similarity of results to the non-stationary conditions of Case 2 suggests that the results for Case 3 are due to systematic error in the direction of the alternative hypothesis. It seems that the robustness of Test 2 to phylogenetic shifts in codon bias is due to generally low power. This finding of low power for Test 2 is consistent with previous analysis of this LRT under different conditions (Zhang et al. 2005).

Other studies (Anisimova and Yang 2007; Yang and dos Reis 2011) found that Test 2 can have good power under certain conditions. Specifically, they found that Test 2 was powerful when the foreground was restricted to a single branch and the level of sequence divergence was optimal (Anisimova and Yang 2007; Yang and dos Reis 2011; Zhang et al. 2005). Note that those results were obtained under ideal conditions, with the simulated distribution of ω being an exact match to the analytical models. Our scenarios are indeed more challenging, as there is a larger gap between the generating models (based on beta distribution and non-stationary codon frequencies) and the analytical models (more simplistic distributions and stationary frequencies). This is intentional, because it is intended to serve as a more realistic portrayal of the gap likely be encountered in the analysis of real data.

Testing for Functional Divergence at the Amino Acid Level

Bivar and FunDi were applied to each of the 18 simulation scenarios. The only null scenario for these methods is Case 1*, as it is the only one where the evolutionary rate is homogeneous among subtrees. Among the three summary statistics employed by Bivar, one (*arsum*) yielded excessively high levels of false positives (34 %) for Case 1*; hence *arsum* will not be further evaluated for power. The other two Bivar statistics (*alrsum* and *abrsum*) provided similar control of the false positive rate, which was only slightly higher than the nominal level of the test (7.5 and 7 %, respectively). The false positive rate produced by FunDi (10 %) was higher than Bivar with *alrsum* and *abrsum*, but substantially lower than Bivar with *arsum* (supplementary Tables S1 and S2).

All the other simulation scenarios (Cases 1a, 1b, 2a–c, 3a–c) were used to investigate power. Because the *alrsum* and *abrsum* statistics were very similar for power (see Supplementary Tables S1 and S2 for all results) hereafter we review only the results for *abrsum*. Table 3 shows only the results for conditions with stationary codon bias. Both tests had moderate but variable power to detect a rate shift (Bivar: 10.5–67.5 %; FunDi: 11–38 %). The Bivar-test had generally higher power than the FunDi-test, with the exception of subcases 2b and 3b. Nevertheless, both approaches had decreased power in the b subcases compared to a and c. The b subcases were more challenging because the shift in selection pressure was not as extreme as the c subcases, and involved a lower proportion of sites than the a subcases. Amino acid models should be less powerful than codon models when synonymous substitutions are not saturated. This was indeed the level of divergence covered by our simulation study, and they did have lower power than the codon-based tests. For instance, branch-site Test 3 detected a shift in constraint in a minimum of 73 % of replicates, which was substantially higher than either amino acid method.

We also applied Bivar and FunDi to simulations with non-stationary codon bias. In these simulations, significance rates increase, though not substantially, when a shift in codon bias is present. In the null scenario (Case 1*), introducing a shift in codon bias caused the false positive rate for Bivar to increase from 7 to 9.5 %, and for FunDi to increase from 10 to 17.5 %. This inflation was systematic throughout all subcases (Table 4). These results indicate that amino acid level tests can be impacted by non-stationary codon bias, which manifests as non-stationary amino acid frequencies (see supplementary Fig. S2).

Other aspects of the substitution process that were not included in the analytical models may also have impacted these tests. Simulations at the codon level in this study were based on a specified distribution of nonsynonymous and synonymous substitutions, without regard for the physiochemical properties of the derived amino acid. Both Bivar and FunDi take into account differential amino acid

Table 3 Performance of tests for shifts in amino acid substitution rates when frequencies are stationary: percent of replicates significant at $\alpha = 0.05$

	Bivar (*abrsum*)	FunDi
Null		
Case 1*	7	10
Small shifts in selective constraints		
Case 1a	54	11
Case 1b	67.5	30.5
Large shifts in selective constraints		
Case 2a	60	38
Case 2b	11.5	24
Case 2c	32.5	22.5
Large shifts with positive selection		
Case 3a	46	30
Case 3b	10.5	27
Case 3c	27.5	22.5

Note for Bivar, results under *arsum* and *alrsum* can be found in the supplementary materials. Case 1* represents homogenous distribution of selection pressure for all branches in the tree, and is the null scenario for both tests. All other cases were employed to evaluate power to detect rate shifts

Table 4 Effect of non-stationary frequencies on tests for shifts in amino acid substitution rates: percent of replicates significant at $\alpha = 0.05$

	Bivar (*abrsum*)	FunDi
Stationary	35.2	23.9
Non-stationary	51.6	36.5

Note for Bivar, results under *arsum* and *alrsum* can be found in the supplementary materials. Here, values are averaged over all cases (a–c). Individual results can be found in supplementary materials

exchangeabilities. Furthermore, they permit one-step changes between amino acid states requiring more than one nucleotide change, and this is not allowed by the codon models. It is possible that the observed impact of non-stationary codon bias is in some way related to this additional gap between the generating and analytical models. To investigate this, we conducted an additional simulation study by generating at the amino acid level under a model that is much closer to assumptions of Bivar and FunDi (see supplementary materials for detailed methods and results). For this study, we simulated a shift in amino acid frequencies equivalent to the shift in codon frequencies used for the main simulation. Under these conditions, Bivar yielded only 2 % false positives and FunDi yielded 6 %. These results indicate that a shift in amino acid frequencies alone will not impact the amino acid level tests as long as the other aspects of models are a good fit to the data.

Analysis of *Prochlorococcus* Core Genome for Functional Divergence Between HL and LL Ecotypes

Genomes of the cyanobacterium *Prochlorococcus marinus* provide an appropriate real-world case of non-stationary sequence evolution associated with functional divergence. *Prochlorococcus* lineages have diverged into two phylogenetically distinct ecotypes, a high-light-adapted ecotype (HL) and a low-light-adapted ecotype (LL). A shift in codon usage is observed between the two ecotypes; the most recently diverged strain has a genomic GC content of 30.8 % while the most deeply branching strain has a genomic GC content of 50.7 % (Fig. 2; Kettler et al. 2007; Rocap et al. 2003). This GC bias, likely a result of mutational bias and rapid evolution (Dufresne et al. 2003), is reflected in biased synonymous codon usage, with the most recently diverged lineage having an effective number of codons (ENC) of 49.0 and the most deeply branching having ENC = 58.2. Because the more recently evolved strains occupy a different habitat, this organism is a prime candidate for functional divergence analysis, yet the non-stationary GC content is a substantial deviation from the analytical models. For this reason, we chose the *Prochlorococcus* core genome as an example of real data analysis under non-stationary conditions.

A 585-gene subset of the core genome was analyzed using branch-site Tests 1 and 2. Because the *Prochlorococcus* gene tree is very asymmetric (as in our simulation study), and sequence divergence tended to be less than in our simulated data sets (average: ~10 compared to 16.2 subst./codon site), we expected that our simulation study would provide a useful framework for interpreting the results of the core genome analysis. Using Test 1, a significant result was observed in 76 % of genes when the entire HL clade was specified as foreground and 90 % of genes when just the branch between ecotypes was specified. However, results from previous studies (Zhang et al. 2005) and our simulations suggest that Test 1 is too liberal, and the false positive rate is further elevated when codon frequencies are non-stationary. For these reasons, this result is likely an overestimate of the number of genes subject to functional divergence. According to the more conservative Test 2, no genes in the subset contain a signal for positive selection when the entire HL clade is specified as foreground and only 24 % when the branch between ecotypes is specified. The latter result might indicate a transient shift in selection pressure. However, because even this test is sensitive to non-stationary frequencies, those results must be treated cautiously.

We attempted to corroborate the codon-level results with amino-acid level analyses. Under Bivar, the *abrsum* measure detected functional divergence in 268 (46 %) genes. The FunDi-test-yielded positive results in 108 (18 %) genes. While results from Bivar and FunDi differ,

Table 5 Results from gene-specific simulations for 10 *Prochlorococcus* genes under two different scenarios: Scenario-1 simulated differences in both the distribution of ω and codon frequencies between HL and LL branches and Scenario-2 simulated differences in only codon frequencies between HL and LL branches

	Scenario 1		Scenario 2	
Gene	Test1	Test2	Test3	Bivar
ABC-transporter	89.5	0	15	3.5
ABC-transporter	75	0	22.5	2
CbbY	45	0	24	1
Diaminopelargonic acid synthase	42.5	0	17.5	3
Oxidoreductase	93	0	23	2
Acetolactate synthase	92.5	0	13	9.5
DNA gyrase modulator	96.5	0.5	16.5	2
UvrD/REP helicase	97.5	0	70.5	0
Methyltransferase	58	0	59.5	1.5
DNA gyrase	93	1	39	0

Results are percent of replicates significant at $\alpha = 0.05$

Note the real sequences for all ten genes had significant *P* values for Tests 1 and 3, but not for Test 2 or Bivar

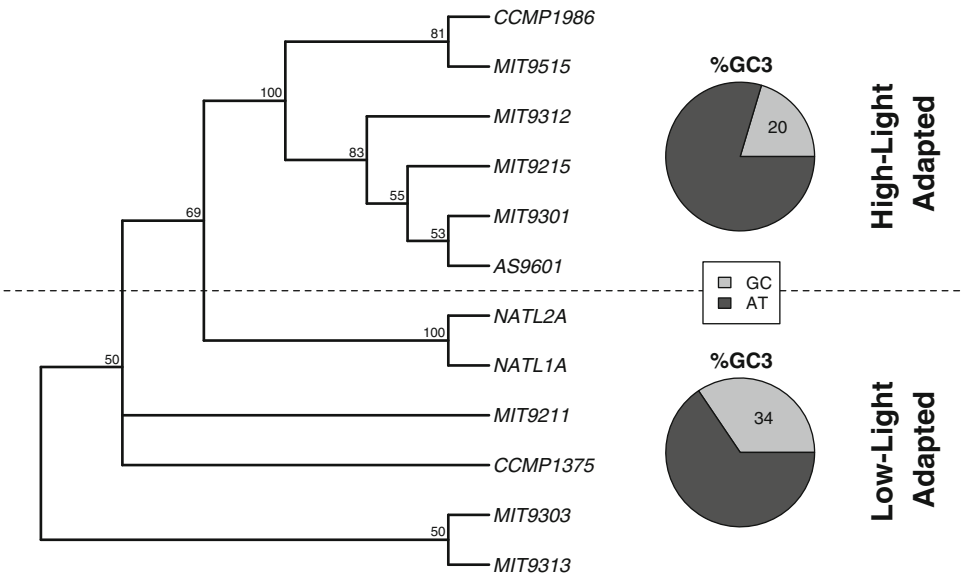

Fig. 2 Consensus tree for 1005 genes from the *Prochlorococcus* core genome. *Node labels* indicate percentage of gene trees with support for that node. The maximum likelihood estimate for each gene tree was estimated using RAxML with a gamma distribution for among-site rate variation. The consensus tree was then calculated using PAUP*. The *dashed line* separates high-light (HL) from low-light (LL)-adapted lineages. Nucleotide composition (%GC3) is shown for the core genome

the simulation study suggests that Bivar has the desirable qualities of moderate power and only small elevation of false positive rates under non-stationary codon bias. To offset potentially elevated false positive rates, we recomputed the results for Bivar under higher stringency by setting $\alpha = 0.01$. Under this condition, 172 (29 %) genes yielded a signal for functional divergence. This value is similar to the codon-level Test 2 results with the branch between ecotypes specified as foreground. However, upon closer analysis, we find that only 50 genes are common to both Bivar and branch-site Test 2.

To further explore the impact of model mispecification on our testing of the *Prochlorococcus* core genome requires gene-specific simulation studies. Due to the computational burden of this approach, we carried out these additional simulations studies for a set of ten genes that showed significant results for Test 1 (Table 5). Choosing genes that exhibited a positive signal for Test 1, we carried out 10 different simulation studies with each being based on gene-specific estimates of the change in codon bias and selection pressure. For each gene, the sequences were separated into HL and LL groups, and a codon model was used to separately estimate the shape of the β distribution under the assumption of no positive selection via ML. Codon frequencies were empirically estimated for the HL and LL sequences as well. A full set of HL- and LL-like sequences was then simulated according to gene-specific branch lengths as estimated from the *Prochlorococcus* genome tree. As positive selection was not permitted, these simulations represent a null scenario for Tests 1 and 2. Application of Test 1 to the simulated HL and LL clades produced an extremely high false positive rate for all ten of the gene-specific scenarios (Table 5: 42–97 %). Clearly, Test 1 is not suitable for real sequences such as *Prochlorococcus* core genome. Taken together with our previous simulations, the empirical results for Test 2 are best explained by a lack of power in this scenario.

We also investigated the performance of codon Test 3 and the Bivar test for detecting a generalized shift in the intensity of selection between the HL and LL clades. Again, we carried out gene-specific simulations, but in this case we employed a joint estimate (HL + LL) of the distribution of selection pressures at sites; i.e., only codon frequencies differed between HL and LL clades in these simulation studies. In these scenarios Test 3 had unacceptable high false positive rates (Table 5: 13–70 %). The rate of false positives was associated with the magnitude of the difference between HL and LL frequencies (supplementary Fig. S3). In all but one of the gene-specific simulations the Bivar test provided adequate control over the false positive rate (Table 5). Reliable inference of functional divergence within the *Prochlorococcus* core genome seems possible under amino-acid level tests such as Bivar, but gene-specific simulations will still be required to validate the findings.

Conclusions

Markovian models for codon evolution typically assume homogeneity (i.e., the rate matrix is the same for all branches of the tree) and stationary composition (i.e., codon frequencies are at equilibrium and unchanging). The branch-site models permit ω to change within the rate matrix for different classes of branches, but not codon frequencies. We found that when evolution is both non-homogenous and non-stationary false conclusions about the

intensity of selection pressure can be obtained. One solution is to implement branch-site models that permit non-stationary codon frequencies. Currently, non-stationary models exist for the nucleotide level (e.g., Yang and Roberts 1995; Galtier and Gouy 1998), but have yet to be developed for the codon level. A major concern is the required increase in parameters. However, this can be avoided by using an approximation based on estimates of nucleotide frequencies at each of the three codon positions (e.g., F3 × 4; Yang 2007). This reduces the burden of the estimation from 61 to just 9 frequency parameters per set of foreground and background lineages. These frequencies could then be estimated using maximum likelihood, similar to what has been done at the nucleotide level (see models N1 and N2 of Yang and Roberts 1995). We are currently developing a set of branch-site models that will accommodate non-stationary codon frequencies. The underlying models of Bivar and FunDi could be likewise extended, although this would require more costly estimation of 20 amino acid frequencies for the different subtrees.

Based on our simulation studies, we provide the following suggestions:

- We concur with Zhang et al. (2005) in recommending that branch-site Test 1 should not be used as a test for positive selection. However, we cannot recommend it as a test for relaxed constraint. It is just as sensitive as branch-site Test 3 to phylogenetic shifts in frequencies, but has lower power to detect smaller shifts in selection intensity.
- Branch-site Test 2 is moderately robust to phylogenetic shifts in codon frequencies, but has low power. It can be used in such cases, but results must be validated with gene-specific simulation studies.
- Branch-site Test 3 is a powerful test for functional divergence, but it is very sensitive to non-stationary codon frequencies. It should not be used when codon frequencies are non-stationary unless accompanied by a gene-specific simulation study.
- When there are phylogenetic shifts in both the distribution of selective constraints and the codon frequencies, the Bivar-test is expected to yield fewer false positives and have generally higher power than FunDi. The *arsum* statistic is not recommended for Bivar.
- Amino acid level tests complement the codon level test, as they can be less sensitive to shifts in codon frequencies.

All the statistical tests, whether based on codon or amino acid models, were negatively affected by errors arising from model misspecification. The problem became most serious when several aspects of the substitution process were miss-specified within the model. Genome-wide scans for positive selection are increasing in popularity; however, it is important that users of these tests understand how violations of model assumptions could impact biological conclusions. Our assessment of these methods in the specific case of the *Prochlorococcus* core genome illustrates that gene-specific simulations are warranted when model assumptions are strongly violated. The striking inconsistency between the results for the real data, taken with the performance of the codon-based tests in the gene-specific simulations, suggests that seeking consensus among methods in genome scans will not necessarily produce reliable results. Although computationally costly, simulation studies motivated by the dataset at hand will allow the output of a model, or test, to be tempered by an understanding of the possible impact of model violations.

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